



cycle4green

LENZING  
PAPIER



## ENVIRONMENTAL FACTS

Cycle4green Liner Recycling – the natural choice

# PAPER RECYCLING IN EUROPE

## Recycling rates in EU:

|                                        |                 |
|----------------------------------------|-----------------|
| Recycling-%, printing papers           | ca. 72 %        |
| Recycled volume, printing paper        | ca. 57 Mio tons |
| Recycling-%, silicone coated paper     | 15 %            |
| Recycled volume, silicone coated paper | 54 000 tons     |

## Impact on CO<sub>2</sub> emissions:

|                          |                                                       |
|--------------------------|-------------------------------------------------------|
| Reduction of emissions : | 1,50 ton CO <sub>2</sub> / ton waste paper (printing) |
| Reduction of emissions : | 2,02 ton CO <sub>2</sub> / ton waste paper (release)  |

## Impact on water consumption:

|                           |                                |
|---------------------------|--------------------------------|
| Reduction of water usage: | 37,5 m <sup>3</sup> / ton pulp |
|---------------------------|--------------------------------|

# RELEASE LINER IN EUROPE

## Release paper used in labelstock:

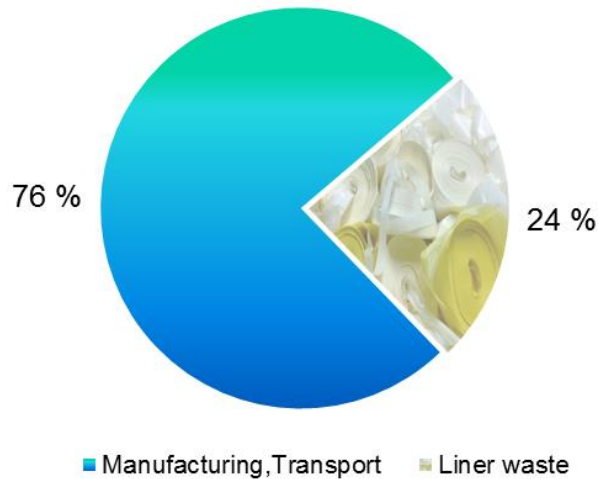
Consumption in EU: 400 000 tons

## Disposal of release liner waste:

|            |      |
|------------|------|
| Recycling: | 15%  |
| Landfill : | 20%  |
| Energy:    | 55 % |
| Overseas:  | 10 % |

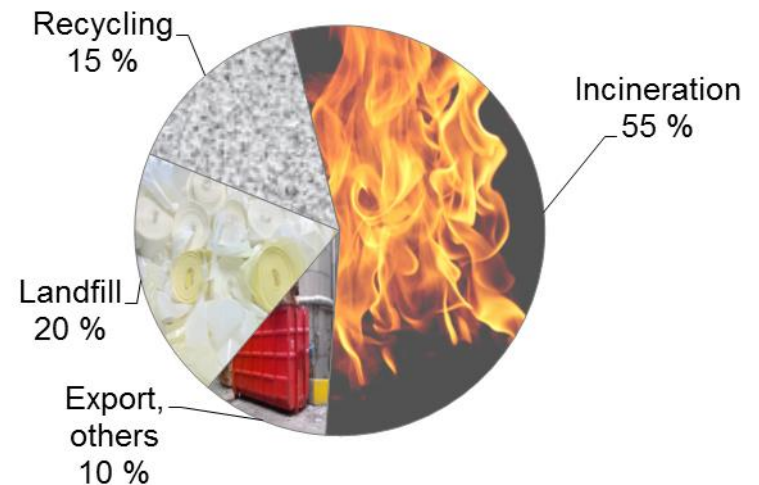
# CARBON FOOTPRINT 1/2

**Carbon footprint - Self-adhesive label material**



- ❑ Fourth of environmental impact originates from liner waste

**End of life - Release Waste Paper**  
% of volume

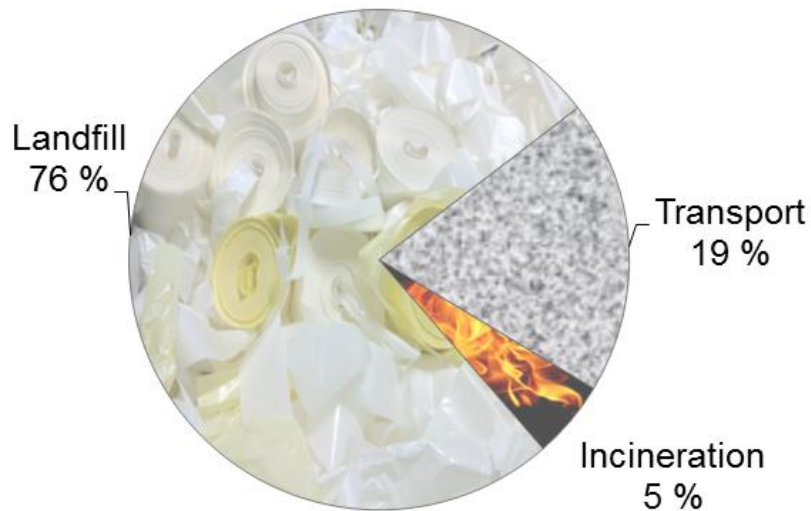


- ❑ Landfilling is becoming less mainly due to more efficient incineration
- ❑ Incineration of organic renewable material is regarded close to carbon neutral providing materials are available locally

# CARBON FOOTPRINT 2/2

## Emissions - Release Waste Paper

% CO<sub>2</sub> ekv.



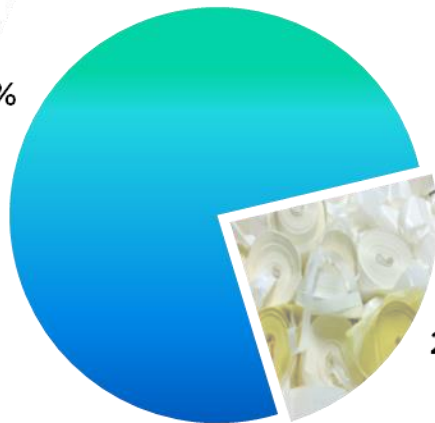
- ❑ Landfilling is ca. 15% of end-of-life disposal but it dominates emissions as organic material releases methane in landfill
- ❑ Emissions from incineration do not take into account emissions from manufacturing of new products needed after disposal

# REDUCTION OF CARBON FOOTPRINT

## Carbon footprint - paper labelstock

Present  
Total **250** g CO<sub>2</sub> / m<sup>2</sup>

76 %



24 %

■ Manufacturing, transport

■ Release waste

Recycled  
Total **240** g CO<sub>2</sub> / m<sup>2</sup>

81 %



19 %

■ Manufacturing, transport

■ Release waste

# CONCLUSIONS

- ✓ **Emissions are dominated by land filling and incineration**
- ✓ **Recycling locally in Europe brings lowest carbon load**
- ✓ **Potential reduction of emissions -15% by C4G Liner Recycling**

## REMARKS:

1. Incineration (energy) is carbon neutral provided that raw material production and incineration is done locally without intercontinental transports
2. Transport: recycling circle 750 km, incineration circle 100 km, overseas 9000 km
3. Calculations based on volumes related to C4G recycling

## REFERENCES:

1. CEPI Guidelines v1.0 2009
2. GHG Protocoll Transport v1.3
3. CEPI Carbon footprints for paper&boards products Sept. 2007, ENV-148-07
4. IPCC (Intergovernmental Panel on Climate Change) GWP (global warming potential)
5. VTT 2009,776 Painotuotteen hiilijalan jälki